MILESTONE 1

Good morning. Our project is a Sensor Data Processing System designed to simulate how farm sensors collect and process environmental data such as soil moisture, temperature, and rainfall.

In Milestone 1, we focused on building the computational foundation of the system. We began by representing real-world parameters such as moisture and temperature as variables in the program.

We then processed multiple moisture readings to compute values such as the average, highest, and lowest moisture levels. This helps in understanding the overall condition of the field.

Next, we classified the moisture readings into categories such as dry, low, optimal, and high. These classifications reflect real agricultural conditions.

Based on this processed data, the system is able to make simple decisions, such as recommending irrigation when moisture is low or issuing warnings when temperature is high.

At this stage, the system follows a procedural approach, focusing on logic and computation.

Building on this foundation, Milestone 2 improves the system structure using object-oriented programming.

MILESTONE 2

In Milestone 2, we extend the system developed in Milestone 1 by introducing object-oriented design.

Instead of using separate variables and logic, we created a Sensor class that encapsulates both the data and behavior of each sensor.

Each sensor object stores its own moisture, temperature, and status, and includes methods to validate and process its readings.

Validation ensures that incorrect sensor values are detected early, preventing faulty data from affecting the system.

The processing method classifies moisture levels and determines actions such as irrigation or drainage checks.

We also simulate multiple sensors using a loop, which represents a real-world farm with several sensors operating simultaneously.

This structure makes the system more organized, scalable, and easier to extend in future milestones.